1.3 PROTONS BEHAVE LIKE TINY COMPASS NEEDLES

From Random Chaos to Alignment in the Main Magnetic Field (B_0)

Protons (spins) are randomly oriented without a magnetic field. When placed in the main magnetic field (B_0), they tend to align with or against the field.

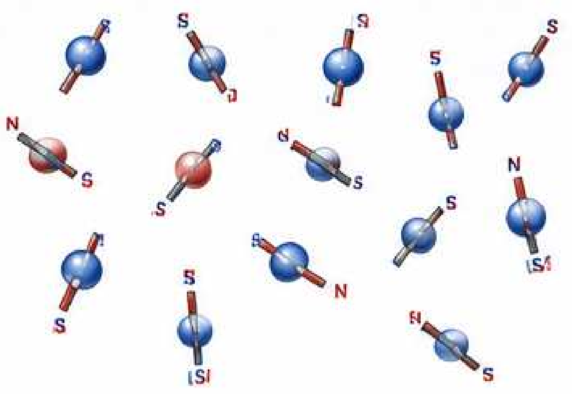
THE CONCEPT

Each proton acts like a tiny bar magnet with a north (N) and south (S) pole. In the absence of a magnetic field, the orientations are completely random. In the presence of B_0, more protons align parallel (low energy) than anti-parallel (high energy).

KEY RESULT

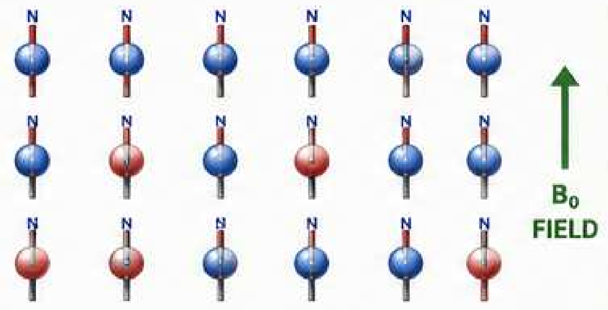
A small excess of protons align with B_0, creating a net magnetization (M_0) along the direction of the field. This net magnetization is the source of the MRI signal.

WITHOUT B_0 FIELD (RANDOM ORIENTATION)



No preferred direction → Net magnetization = 0

INSIDE B_0 FIELD (PARTIAL ALIGNMENT)

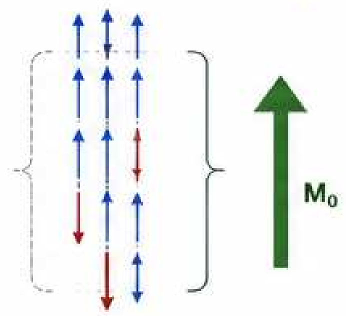


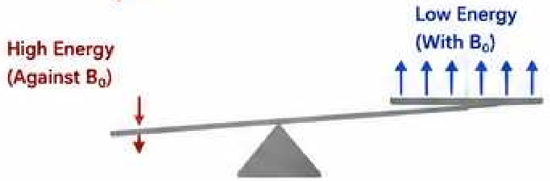
Aligned with B_0 (Low energy)

Against B_0 (High energy)

More align with B_0 than against → Net magnetization (M_0)

NET MAGNETIZATION (M_0)





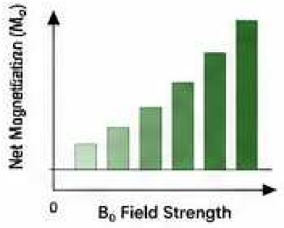
M_0 is the vector sum of all proton magnetic moments.

Direction:
Along the B_0 field (longitudinal axis)
Symbol: M_0
Units: Ampere-turns per meter (A/m)

Important:
The difference is small (~ few parts per million) but critical for MRI!

ENGINEERING INSIGHT

The stronger the B_0 field, the greater the alignment and the larger the net magnetization (M_0).
This leads to a stronger MRI signal.



FIELD SERVICE TIP

Always verify main magnet field strength and homogeneity. Poor field quality reduces proton alignment and degrades image SNR.

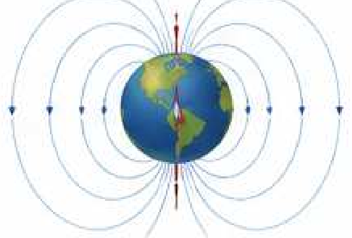
COMMON MISCONCEPTION

Protons do not line up 100% with B_0.
There is always a small population aligned against the field (high energy).

REAL-WORLD ANALOGY

Like compass needles:

- Without Earth's magnetic field → random
- In Earth's magnetic field → most point North (some still point South)



KEY TAKEAWAY In the B_0 field, more protons align with the field than against it. This creates a net magnetization (M_0) along the longitudinal axis, which is the foundation for all MRI signal generation.

NEXT >> Next: Section 1.4 – Precession: Spinning Like a Top
We'll see how these aligned protons start to precess around B_0.